Rongzhe Wei

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Education

Georgia Institute of Technology 2021–Present | Expected Dec. 2026
Ph.D. in Machine Learning, School of Electrical and Computer Engineering
Advisor: Prof. Pan Li | GPA: 4.0/4.0

Xi'an Jiaotong University 2017–2021
B.S. in Mathematics, Honors Math Program, Qian Xuesen College
GPA: 3.89/4.0 | Ministry of Education Honors Math Program

Research Interests

AI Safety & Alignment: Red-Teaming, Multi-Turn Jailbreaks & Defenses, Safety Guardrails, Supervised Fine-Tuning (SFT) & Reinforcement Learning (RL) for Safety.
Trustworthy AI: Machine Unlearning & Knowledge Editing, Differential Privacy.
World Models & Physical AI: Representation Learning for Planning, Latent Dynamics Prediction.
Agentic AI: Long-Horizon Planning, Tool Use, Search & Exploration, Agentic Attack & Defense.
Graph-Structured Learning.

Research Highlights

GCoM Lab, Georgia Tech *Algorithms, empirical studies, and theory*

Adaptive red-teaming of frontier language models

CKA-Agent | ICML 2026 [1]

- Developed an automated red-teaming agent that probes **distributed malicious intent** through adaptive tree search over correlated knowledge, exposing failures missed by prompt-level safety checks.
- Achieved **95.1–98.8% full attack success** across HarmBench and StrongREJECT in the reported four-model evaluation, including Gemini-2.5-Pro and Claude-Haiku-4.5.

Training multi-turn safety guardrails with SFT and RL

TurnGate | Preprint / Under Review, 2026 [17]

- Formulated response-aware defense as a **sequential stopping problem**; developed MTID with first harm-enabling-turn labels and matched benign hard negatives, then trained a Qwen3-4B guardrail with SFT and multi-turn RL using intervention-timing rewards.
- On held-out MTID trajectories, RL reduced missed interventions from **37.9% to 17.7%** versus the SFT initialization while preserving comparable benign pass-through (83.4% vs. 84.0%).

State-aware attack and defense for tool-using agents

SEAD | Preprint / Under Review, 2026 [20]

- Co-developed a partially observed state-control framework with executable outcome checks; the SAGE defender inspects relevant environment state before permitting tool actions.
- In online evaluation, SAGE reduced executable attack success from **48% to 4%**. On recorded trajectories, it preserved 95.79% of benign paths and intercepted 92.73% of harmful paths by the harm-enabling boundary.

Evaluating knowledge removal and privacy in LLM unlearning

Knowledge correlation & minority-aware privacy audits | NeurIPS 2025; ICML 2025 [4], [6]

- Developed confidence-aware, knowledge-graph evaluations that test whether supposedly forgotten facts remain recoverable through inference; exposed overestimation of unlearning by direct-recall tests.
- Designed minority-aware privacy audits using PII canaries and membership inference, revealing **at least 20% greater privacy leakage in most evaluated settings** for the minority subsets studied.

Research Highlights (continued)

Deep unlearning across knowledge paths and MoE experts

Deep unlearning: Preprint / Under Review, 2026 [18] | GRIP: COLM 2026 [2]

- **Led and mentored multiple research interns** on LLM unlearning. As corresponding author, developed a framework that searches supporting reasoning paths and uses **graph minimum cuts** to remove recovery routes while preserving unrelated knowledge.
- Co-developed GRIP to constrain expert routing and prevent unlearning shortcuts in MoE models; the reported evaluation reduced white-box knowledge recovery from **11% to 3%**.

Efficient, routing-stable knowledge editing for MoE LLMs

MoEEdit | ICLR 2026 [3]

- Co-developed MoEEdit, using **per-expert null-space projections** to preserve router inputs during factual updates and block coordinate descent for efficient optimization; improved editing efficacy and generalization while preserving specificity and routing stability.

Representation sufficiency in world models for planning

World-model theory & controlled experiments | Preprint / Under Review, 2026 [19]

- Formalized a hierarchy of **mechanism, response, and decision sufficiency**, distinguishing information needed to predict outcomes, discover candidate actions, and select a low-regret action.
- Tested these distinctions in collision systems, nonlinear dynamics, and robotic planning; motivated a modular design that separates query-guided action proposals from reusable physical predictions.

Formal privacy guarantees for iterative learning

Privacy amplification by iteration & diffusion models | NeurIPS 2024; TMLR 2024 [10], [11]

- Developed ∞-Wasserstein distance tracking within shifted Rényi divergence analysis, tightening non-divergent privacy bounds and replacing a restrictive global-diameter bound. Extended privacy amplification by iteration to Laplace-noise graph diffusion.
- Translated the analysis into edge-private Personalized PageRank with improved utility under stringent privacy budgets. Derived per-instance privacy bounds for discrete denoising diffusion models, characterizing how leakage depends on the data distribution and denoising process.

Statistical foundations of graph neural networks

Bayesian inference & generalization | NeurIPS 2022 [14]

- Derived nonlinear graph aggregation from Bayesian inference under a contextual stochastic block model; **proved when nonlinear aggregation improves over linear alternatives**, including under attribute distribution shift, and validated the predictions on synthetic and real-world graphs.

Industry Experience

Amazon AI Research Intern, Seattle May–Aug. 2025

Long-horizon tool-use evaluation and search [15]

- Built **SLATE**, a 1,000-tool benchmark for automated plan-level evaluation, and developed **Entropy-Guided Branching** to focus search on uncertain decisions. Achieved **54% execution success** with Claude-Sonnet-4, versus 29.3% for ReAct and 36.5% for LATS.

J.P. Morgan Chase & Co. AI Research Intern, New York June–Aug. 2023

Research on summary-statistic privacy and graph data generation using margin-relaxed Schrödinger bridges. Mentors: Eleonora Kreačić and Vamsi K. Potluru. [16]

Research Leadership & Funding

- **Amazon Research Award (2026).** Lead writer of the successful proposal *Structure-Guided Agent Safety* for Prof. Pan Li, the award recipient.
- **NAIRR Pilot Award (Feb. 2026), proposal lead writer.** *Structured-Knowledge-Guided Agentic LLM Jailbreaking and Defense.*
- **Invited talks: Google Research** (Feb. 2026) **and IBM Research** (Jan. 2026). *From Atomic Facts to Structured Internal Knowledge: Rethinking Unlearning and Jailbreaking in LLMs.*

Publications

Full author lists; **bold** indicates my name. * Equal contribution. † Corresponding author.

Conference & Journal Papers

[1] **The Trojan Knowledge: Bypassing Commercial LLM Guardrails via Harmless Prompt Weaving and Adaptive Tree Search**
Rongzhe Wei[*], Peizhi Niu[*], Xinjie Shen[*], Tony Tu, Yifan Li, Ruihan Wu, Eli Chien, Pin-Yu Chen, Olgica Milenkovic, Pan Li.
ICML 2026 Paper Code

[2] **GRIP: Algorithm-Agnostic Machine Unlearning for Mixture-of-Experts via Geometric Router Constraints**
Andy Zhu, **Rongzhe Wei**, Yupu Gu, Pan Li.
COLM 2026 Paper Code

[3] **MoEEdit: Efficient and Routing-Stable Knowledge Editing for Mixture-of-Experts LLMs**
Yupu Gu, **Rongzhe Wei**, Andy Zhu, Pan Li.
ICLR 2026 Paper

[4] **Do LLMs Really Forget? Evaluating Unlearning with Knowledge Correlation and Confidence Awareness**
Rongzhe Wei[*], Peizhi Niu[*], Hans Hao-Hsun Hsu, Ruihan Wu, Haoteng Yin, Mohsen Ghassemi, Yifan Li, Vamsi K. Potluru, Eli Chien, Kamalika Chaudhuri, Olgica Milenkovic, Pan Li.
NeurIPS 2025 Paper Code

[5] **Differentially Private Relational Learning with Entity-level Privacy Guarantees**
Yinan Huang[*], Haoteng Yin[*], Eli Chien, **Rongzhe Wei**, Pan Li.
NeurIPS 2025 Paper

[6] **Underestimated Privacy Risks for Minority Populations in Large Language Model Unlearning**
Rongzhe Wei, Mufei Li, Mohsen Ghassemi, Eleonora Kreačić, Yifan Li, Xiang Yue, Bo Li, Vamsi K. Potluru, Pan Li, Eli Chien.
ICML 2025 Paper Code

[7] **Generalization Principles for Inference over Text-Attributed Graphs with Large Language Models**
Haoyu Wang, Shikun Liu, **Rongzhe Wei**, Pan Li.
ICML 2025 Paper

[8] **Privately Learning from Graphs with Applications in Fine-tuning Large Language Models**
Haoteng Yin, **Rongzhe Wei**, Eli Chien, Pan Li.
COLM 2025 Paper

[9] **Towards Universal Debiasing for Language Models-based Tabular Data Generation**
Tianchun Li, Tianci Liu, Xingchen Wang, **Rongzhe Wei**, Pan Li, Lu Su, Jing Gao.
Findings of EMNLP 2025 Paper

[10] **Differentially Private Graph Diffusion with Applications in Personalized PageRanks**
Rongzhe Wei, Eli Chien, Pan Li.
NeurIPS 2024 Paper

[11] **On the Inherent Privacy Properties of Discrete Denoising Diffusion Models**
Rongzhe Wei, Eleonora Kreačić, Haoyu Wang, Haoteng Yin, Eli Chien, Vamsi K. Potluru, Pan Li.
TMLR 2024 Presented at ICLR 2025 Paper

[12] **Learning Scalable Structural Representations for Link Prediction with Bloom Signatures**
Tianyi Zhang[*], Haoteng Yin[*], **Rongzhe Wei**, Pan Li, Anshumali Shrivastava.
WWW 2024 Paper

[13] **SLA^{2}P: Self-supervised Anomaly Detection with Adversarial Perturbation**
Yizhou Wang, Can Qin, **Rongzhe Wei**, Yi Xu, Yue Bai, Yun Fu.
IEEE TKDE 2024 Paper

[14] **Understanding Non-linearity in Graph Neural Networks from the Perspective of Bayesian Inference**
Rongzhe Wei, Haoteng Yin, Junteng Jia, Austin R. Benson, Pan Li.
NeurIPS 2022 Paper

Publications (continued)

Workshop Papers

[15] **Long-Horizon Plan Execution in Large Tool Spaces through Entropy-Guided Branching**
Rongzhe Wei, Ge Shi, Min Cheng, Na Zhang, Pan Li, Sarthak Ghosh, Vaibhav Gorde, Leman Akoglu.
NeurIPS 2026 AABA4ET Workshop Paper

[16] **Guarding Multiple Secrets: Enhanced Summary Statistic Privacy for Data Sharing**
Shuaiqi Wang, **Rongzhe Wei**, Mohsen Ghassemi, Eleonora Kreačić, Vamsi K. Potluru.
ICLR 2024 Privacy in Machine Learning Workshop Paper

Preprints / Under Review

[17] **One Turn Too Late: Learning When to Intervene Against Multi-Turn Malicious Intent**
Xinjie Shen*, **Rongzhe Wei***, Peizhi Niu*, Haoyu Wang, Ruihan Wu, Eli Chien, Bo Li, Pin-Yu Chen, Pan Li.
Preprint / Under Review, 2026 Paper Code

[18] **Making LLMs Truly Forget: Deep Unlearning by Searching, Selecting, and Severing Knowledge Paths**
Jialu Wang, Peizhi Niu, Haoteng Yin, Hans Hao-Hsun Hsu, Pan Li, **Rongzhe Wei**†.
Preprint / Under Review, 2026 Paper

[19] **What Must a World Model Distinguish for Planning?**
Rongzhe Wei, Hans Hao-Hsun Hsu, Peizhi Niu, Yifan Li, Pan Li.
Preprint / Under Review, 2026 Paper

[20] **SEAD: A State-Based Perspective on Attack and Defense in Tool-Using Agents**
Xinjie Shen*, Junran Wang*, **Rongzhe Wei**, Pan Li.
Preprint / Under Review, 2026 Paper Code

[21] **Beyond the Sequence: Graph Learning as the Blueprint for Trustworthy Large Language Models**
Mufei Li*, Shikun Liu*, Xinnan Dai*, **Rongzhe Wei***, Haoyu Wang, Xinjie Shen, Siqi Miao, Jiliang Tang, Pan Li.
Under Review

[22] **Learning and Steering Protein Flexibility for Generative Protein Design**
Tony Tu, Jiaqi Luo, Ziang Li, Kerr Ding, Yinan Huang, **Rongzhe Wei**, Xinjie Shen, Pan Li, Yunan Luo.
Under Review

Honors & Awards

Georgia Tech CSIP Outstanding Research Award	2025
Lambda Research Grant; OpenAI Researcher Access Program Grant	2025
NeurIPS Travel Award	2022
IEEE BigData Student Award	2019
First Prize Zhufeng Scholarship; XJTU Outstanding Student Award / First-class Scholarship	2017–2020

Professional Service & Technical Skills

Conference reviewing: NeurIPS 2022–2026; ICML 2024–2026; ICLR 2025–2027; AISTATS 2023–2024; ISIT 2025; AAAI 2024; LoG 2022–2024.

Journal reviewing: TMLR; Computers and Mathematics with Applications.

Teaching: Conversational AI; Convex Optimization; Introduction to Probability & Statistics.

Research stack: Python, PyTorch, Hugging Face Transformers, vLLM, DeepSpeed, AutoAWQ, AutoGPTQ.

Additional: C#, MATLAB, SQL, LaTeX.

Languages: Mandarin (native), English (fluent).